

JACOB COOK

Cyber Operations Analyst II | M365 / Defender XDR · Sentinel KQL · CMMC 2.0 · NIST 800-171 · MITRE ATT&CK · Vulnerability Mgmt

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Detection-focused security analyst with hands-on experience triaging escalated alerts, authoring Sigma rules mapped to MITRE ATT&CK, and tuning detections to suppress false positives. Lab work spans Wazuh SIEM, Microsoft Sentinel/Defender concepts, KQL queries, and Entra ID identity work directly relevant to M365 E5 environments. Comfortable with CMMC 2.0 / NIST 800-171 framework concepts. US citizen. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress.

TECHNICAL SKILLS

Detection & IR: Escalated Alert Triage, Lateral Movement / Privilege Escalation Analysis, Live Incident Response (Containment / Eradication / Recovery), MITRE ATT&CK, Zero Trust Concepts, Threat Hunting, Sigma Rule Authoring, RCA & Timeline Reporting
Microsoft Stack: Microsoft Sentinel (familiar), Defender XDR Concepts (Endpoint, O365, Cloud Apps, Identity), Entra ID, KQL (familiar), Workbooks & Analytics Rules
Vuln & Compliance: Vulnerability Lifecycle, CVSS Triage & Prioritization, Patch Cycles, CMMC 2.0 / NIST 800-171 (familiar), FedRAMP Concepts, Audit Evidence
Tooling & Dev: Wazuh SIEM (build & admin), Python, Bash, SQL, FastAPI, Git, AI-Assisted Investigation

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code 2026

- Authored 6 Sigma-format detection rules for Intune (Microsoft endpoint mgmt) identity abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation.
- Documented 3 benign scenarios per rule (18 total) so analysts can triage M365-style identity activity confidently without escalating known-good behavior.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py).

Blue Team SOC Monitoring Lab: Wazuh SIEM 2026

- Built a Wazuh SIEM stack from scratch and validated 12 detection patterns against 242 simulated auth-style events with 100% precision after tuning — the same alert-threshold-and-rule tuning work expected at COA 2 level.
- Reduced false-positive volume by 100% by tuning out 79 benign-only matches (cron jobs, service accounts, scheduled deployments) without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

SOAR-lite: Incident Response Orchestrator 2026

- Built end-to-end alert automation: webhook ingestion, playbook routing, IP blocking, Slack notification, and incident write-up — the same response-playbook pattern Sentinel automation rules use for repeatable customer-tenant containment.
- Eliminated four manual triage steps and reduced triage time by over 60% in lab testing.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py).

Phishing Analysis Lab 2026

- Triaged 5 real-world phishing samples end-to-end with header inspection, SPF/DKIM/DMARC validation, IOC extraction, and attacker infrastructure mapping — 100% accurate verdicts against ground truth.
- Delivered structured incident write-ups in the format expected for high-touch customer briefings during active phishing investigations.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6) 2024

- Engineered RBAC and Azure AD / Entra ID SSO controls — direct M365-stack identity work — that enabled executive-level adoption with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response 2019 - 2025

- Maintained 99.5% system uptime by triaging telemetry alerts on rotating shifts — direct continuous-monitoring discipline transferable to multi-tenant SOC operations.
- Resolved 200+ anomalies through structured RCA and timeline analysis, producing tracked documentation that reduced repeat incidents by an estimated 30%.
- Reduced mean time to recovery by an estimated 20% via cross-team coordination, escalation, and post-incident reporting for technical and non-technical stakeholders.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)